ChE 381 – Chemical Engineering Laboratory I- Fall 2015

Student Learning Outcomes 1, 3 (written and oral), 4, 5, 6, 7

Rubric for assessment for Pre-lab Report, Final Lab Report, Final Oral Presentation, Team Skills Assessment, SACHE Safety Tutorial

RESULTS BY PERFORMANCE INDICATOR: Prelab, Final Lab Reports											
Rubric item (PI)	1	2	3	4	5	6	7	8	9	10	11
Mapped SLO	1,3	3,6	3,4,7	3	1,3,6	1,3,6	3,6	3,7	1,3,6	1,3,6	1,3,6
Average	2.66	2.82	2.92	2.66	2.42	2.34	2.34	3.00	3.00	3.00	2.81
Std Dev	0.48	0.39	0.27	0.48	0.50	0.63	0.48	0.00	0.00	0.00	0.60

RESULTS BY PERFORMANCE INDICATOR: Oral Presentation				
Rubric item	12	13	14	15
Mapped SLO	3	3	3	3
Average	2.69	2.48	2.70	2.65
Std Dev	0.46	0.41	0.38	0.32

RESULTS BY PERFORMANCE INDICATOR: Team Assessmen			
Rubric item	16	17	18
Mapped SLO	5	5	5
Average	2.74	2.76	2.76
Std Dev	0.48	0.50	0.53

RESULTS BY PERFORMANCE INDICATOR: Safety	
Rubric item	19
Mapped SLO	4
Average	3.00
Std Dev	0.00

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS							
	1	3	3	4	5	6	7
		Written	Oral				
Average Score	2.70	2.72	2.63	2.96	2.75	2.68	2.96
Standard dev.	0.28	0.26	0.10	0.06	0.01	0.30	0.06

Average scores showed most students were very strong in experimental preparation, discussion of results, lab and industrial safety, sample calculations, data tabulation. A large majority were satisfactory in analyzing the experimental data and drawing conclusions. Most students were also strong in teamwork/team skills.

Although the students show satisfactory attainment of PI6, error analysis was not as strong. To address this issue, supplemental information will be given to the students and will be made readily available and emphasized when they are performing the experiments. Error analysis topics include; determining the degree of uncertainty (systematic erros, instrumental uncertaintly, random errors), accuracy, precision/reproducibility, and statistical analysis.

Assessment Rubric for Lab reports (pre-lab and final lab report sections combined)

Report Section, Performance Indicator (PI)	**SLO's**	**(1) Unsatisfactory**	**(2) Developing**	**(3) Satisfactory**
1.Introduction	SLO 1, 3	Main aspects of the lab experiments are not addressed; no experimental objective, experimental description or discussion of industrial applications, poor grammar and usage	Several main aspects of lab experiment not addressed or not stated in a clear and concise manner.	Main aspects of lab experiment addressed and clearly stated; minor grammar and usage issues.
2. Experimental: Apparatus, Materials & Supplies, Procedure	SLO 3, 6	Incomplete process diagram, missing materials/supplies list, incomplete or vague experimental procedure	Adequate process diagram, material/ supply list, experimental procedure; several problems or mistakes	Complete process diagram, material/ supply list, experimental procedure with minor problems
3. Project Safety Evaluation	SLO 3, 4, 7	Missing or lacking an understanding of major/ minor safety hazards and safe operating conditions	Safety understanding, missing some major safety hazards	Safety understanding and important safety hazards, missing some minor hazards
4. Executive Summary	SLO 3	Illegible or incoherent summary of conclusions, poor grammar and formatting	A few problems with summary of conclusions, legibility or neatness	Minor problems with summary of conclusions, grammar, formatting, neatness
5. Results	SLO 1, 3, 6	Drawings, graphs and tables unacceptable, confusing, or missing	Several deficiencies in drawings, graphs and tables. Figures and tables not referred to in text.	Only a few deficiencies with drawings, graphs and tables
6. Discussion	SLO 1, 3, 6	Main points and trends from results are missing or not discussed clearly; lack of understanding of the engineering reasoning behind the analysis	Several main points and trends are missing or not discussed clearly; some understanding of engineering reasoning behind analysis	Only a few points or trends are missing or not discussed clearly; a substantial understanding of the engineering reasoning behind analysis
7. Conclusions	SLO 3, 6	No significant conclusions drawn, Limitations of results and future work not discussed.	Cursory conclusions drawn from data and calculations. Some discussion of limitation of results and future work.	Valid conclusions drawn from data and calculations. Future work addressed in relation to limitations or uncertainties.
8. References	SLO 3, 7	Seeks no extra information other than what is provided by instructor	Seeks information from a few sources, mainly from the textbook or instructor	Seeks information on problems from multiple sources

9. Data Tabulation	SLO 1, 3, 6	Experimental data tables and graphs unacceptable or missing	Several deficiencies with data tables and graphs	Only a few deficiencies in data tables and graphs
10. Sample Calculations	SLO 1, 3, 6	Calculations are incorrect, presented in a confusing way and do not link with specific experimental results	Some calculations are incorrect or presented in a confusing way	One or two calculations are incorrect or presented in a confusing way
11.Error Analysis	SLO 1, 3, 6	Cursory and qualitative description of errors	Qualitative analysis of error with some calculations	Quantitative and qualitative error analysis, missing some minor calculations

Assessment Grading for Lab reports (pre-lab and final lab report sections combined)

Report Section, PI	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Introduction	SLO 1, 3	0-2	3	4-5
2. Experimental: Apparatus, Materials & Supplies, Procedure	SLO 3, 6	0-11	12-15	16-20
3. Project Safety Evaluation	SLO 3, 4, 7	0-5	6-7	8-10
4. Executive Summary	SLO 3	0-5	6-7	8-10
5. Results	SLO 1, 3, 6	0-8	9-11	12-15
6. Discussion	SLO 1, 3, 6	0-8	9-11	12-15
7. Conclusions	SLO 3, 6	0-6	7-8	9-10
8. References	SLO 3, 7	0-2	3	4-5
9. Data Tabulation	SLO 1, 3, 6	0-8	9-11	12-15
10. Sample Calculations	SLO 1, 3, 6	0-8	9-11	12-15
11.Error Analysis	SLO 1, 3, 6	0-8	9-11	12-15

Assessment Rubric for Oral Presentation

Performance Indicator (PI)	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
12. Organization	SLO 3	Talk is poorly organized, e.g. no clear introduction or summary of talk is presented; time is too short or too long	. Presents key elements of an oral presentation adequately, but "tell them" not clearly applied; time is slightly too long or too short	Plans and delivers an oral presentation effectively; applies the principle of "(tell them)3" --well organized; time is just right

13. Delivery	SLO 3	"Major difficulties with the mechanical aspects of the presentation · No eye contact · Difficult to hear or understand speaking · Reads from prepared script · Blocks the screen · Distracting nervous habits (um, ah, clicking pointer, etc.)"	"Has some minor difficulties with the mechanical aspects of the presentation · Eye contact is sporadic · Occasionally difficult to hear or understand speaking · Overuses prompts or does not use prompts enough- occasionally stumbles or loses place; appears to have memorized presentation	Presents well mechanically: · Makes eye contact · Can be easily heard · Speaks comfortably with minimal prompts (notecards) · Does not block screen · No distracting nervous habits
14. Visual Aids	SLO 3	Multiple slides are unclear or incomprehensible	Visual aides have minor errors or are not always clearly visible	Uses visual aides effectively
15. Content	SLO 3	Omits key results during presentation	Presentation is lacking some details or technical content	Presentation has enough detail appropriate and technical content for the time constraint and the audience

Oral Presentation Evaluation Form

FINAL LAB PRESENTATION
EVALUATION FORM
CHE381 CHEMICAL ENGINEERING LABORATORY

TEAM #: LAB EXPERIMENT: ______________________

COMMENTS:

1. Organization (Use of time, overview of presentation, summary)
2. Presentation Delivery (Eye contact, voice, answering questions)
3. Visual Aids (Neatness, use of space on slides, extras (animation, etc))
4. Content (Level of information, discussion, analysis)

Presentation Grade: (100 points max):

Assessment Rubric for Team Member Skills

Performance Indicator (PI)	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
16. Quality Work	SLO 5	Has a poor grasp of chemical engineering principles and poor writing skills	Has some knowledge of chemical engineering principles and writing skills	Has good knowledge of chemical engineering principles and writing skills
17. Teamwork	SLO 5	Hides in the background; only participates if strongly encouraged Routinely fails to prepare any plans, does a minimal amount of task work	Occasionally plans and carries out lab tasks. Occasionally takes charge when not in the position to lead	Demonstrates the ability to assume a designated role in the group, plan tasks, carry out tasks and contributes to success of team
18. On-time tasks	SLO 5	Consistently hands in task late to the team	Occasionally is late in handing in tasks	Hands in tasks (lab report sections) on time to team

Team Member Assessment Evaluation Form:

CHE 381 TEAM MEMBER ASSESSSMENT SHEET

Submitted by:________________________________ Team #______ Date:

The team member review form is given below and shall be submitted to Professor Zdunek by email on or before 5PM Monday December 7th.

Please be fair and honest in your assessment. I will take the team grading into consideration when preparing the Chemical Engineering Laboratory grades-

- A lack of positive, meaningful team participation can affect your individual grade.
- If you do not submit a team member review form on-time, your grade will also be impacted.

Thank you!

TEAM MEMBER YOU ARE ASSESSING:___________________________________.
1 = never, 5 = always

	1	2	3	4	5
Did he/she submit quality, complete work for assigned actions					
Did he/she participate in a cooperative and respectful manner					
Did he/she meet promised dates for assigned actions					

TEAM MEMBER YOU ARE ASSESSING:___________________________________.
1 = never, 5 = always

	1	2	3	4	5
Did he/she submit quality, complete work for assigned actions					
Did he/she participate in a cooperative and respectful manner					
Did he/she meet promised dates for assigned actions					

-More forms on next page-

Assessment Rubric for Safety Assignments

Performance Indicator (PI)	**SLO's**	**(1) Unsatisfactory**	**(2) Developing**	**(3) Satisfactory**
Understanding industrial safety and practices	SLO 4	Missing or lacking an understanding of industrial safety practices	Has some understanding of industrial hazards and safety practices	Has a good understanding of industrial hazards and safety practices Example: Passed a safety tutorial (SACHE)
Understanding of lab safety and safety evaluations	SLO 4	Missing or lacking an understanding of major/ minor safety hazards and safe operating conditions	Missing some major/minor aspects of laboratory safety practices	Satisfactory understanding of laboratory safety covered. Example: Understands a laboratory Chemical Hygiene Plan

Safety Assignment: SACHE Tutorial, "Chemical Reactivity Hazards"

SACHE-AICHE safety tutorial website

AICHE (American Institute of Chemical Engineers) has created a website with various safety tutorials covering process safety, risk assessment and chemical reactivity hazards.

ASSIGNMENT: Complete the tutorial titled "Chemical Reactivity Hazards" in the website. Given below is the procedure on how to access the website.

NOTE: You will be asked for either your AICHE member ID or an AICHE user ID. If you do not have either one, you have the option of joining AICHE as a student member (free) or signing up for access to the AICHE website. I SUGGEST YOU DO THIS EARLY IN THE SEMESTER IN CASE THERE ARE PROBLEMS WITH SIGNING IN TO THE WEBSITE. THERE WILL BE NO EXTENSIONS FOR LATE COMPLETIONS.

GRADING: You must email proof (Certificate issued by SACHE) that you have completed the tutorial and answered the questions for full credit.

DUE: Wednesday November 26, 2014 at 1:00 PM

WEBSITE PROCEDURE:

1. Go to www.sache.org and click on "Login" in the Menu on the left hand side of the page.
2. Use the Alternate Login: University Student Login
    a. Select University: University of Illinois at Chicago
    b. Type the Password: chemeisgreat (no cap or spaces)
3. Click on "Student Certificate Program" to get into the tutorial section. Pick the safety tutorial:
    a. Chemical Reactivity Hazard

You will then need to type in your AICHE member ID/password or AICHE user ID /password in order to gain access to the specific tutorial. If you do not have either one- sign up to be a student member or to be a AICHE website user.

4. Print out Certificate that you completed the tutorial and email to TA for credit.

Good Luck!

Professor Zdunek